



## Finals Section 4 - Case Study Information Pack

### Section 4: Case Study – Cakes and Onions

**Relates to Questions 1 - 20**

**120 Marks available in this Section**

This section will be objectively assessed based on a combination of multi-choice questions (Q1 to Q17), free-field response questions (Q18 to Q20) and the dynamic capability of your model (all questions). It will be marked out of 120.

For each question either circle or write your answer as required. Please also include a reference for the sheet and cell address of each answer where noted in the answer tables.

#### INTRODUCTION

1. You are working for a newly established mega-company called ModelOff Vessels (“MOV”) that plans to produce high-end ships (vessels). MOV has already acquired its initial fixed assets and raised a range of financing (both debt and equity). MOV will begin taking orders for its vessels next year (2017).
2. MOV will sell five different types of vessel: Platinum class, Ruby class, Emerald class, Gold class and Sapphire class.
3. There are three different volume forecast scenarios for the vessel sales. These scenarios are called Base Case, Declining Case and Cyclical Case. Your model must be able to produce outputs for all three cases (either through a dynamic toggle/switch/cell or by calculating all three scenarios at the same time in the model). For every question, you will have to provide an answer for all three scenarios. For each question, the Base Case answer is worth 3 marks, the Declining Case answer is worth 2 marks and the Cyclical Case answer is worth 1 mark.
4. You will need to develop a model to forecast MOV’s business operations including an integrated Income Statement, Balance Sheet and Cash Flow Statement. The model will have annual periodicity and cover 30 years from 1 January 2017 until 31 December 2046.
5. Assume that all cash flows in the Model occur on the final day of the period.
6. The key model inputs (discussed below) have been provided to you in the Excel file titled ‘MO16 Finals - Sec 4 – Cakes and Onions.xlsx’ (the “Inputs Workbook”).
7. You should not round your calculations at any point in the model.
8. The multiple choice answers presented to you may be rounded. Answer to the level of precision displayed in the answers (i.e. if your model calculation gave a result of 53.68292 and the options on the answer sheet are 53.6 or 53.7, you should select 53.7).
9. To assist with formatting, a legend is provided in the ‘Cover’ sheet with most formats you will wish to use. A Custom Style for each format in the legend has also been added to the workbook.



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### INFLATION

1. Where amounts are to be inflated they are given in 2016 prices and inflation should be applied on 1 January of each subsequent year (beginning 1 January 2017).
2. Price inflation applies to the price that customers pay for the vessels.
3. Cost inflation applies to all 'other operating expenses' (head office costs, insurance, government fees and other expenses) and to all capital expenditures. It is not applicable to COGS (see more information on COGS below).
4. The annual inflation percentages are provided in the Inputs Workbook.

### OPERATIONS

5. The forecast number of vessels sold in each year is provided in the Inputs Workbook.
6. MOV has an opening retained earnings balance of zero.
7. Production of a vessel begins in the year the order is placed. Each vessel required a different number of years to produce (Platinum class vessels require 5 years, Ruby and Emerald class vessels require 4 years, Gold and Sapphire class vessels require 3 years).
8. You may assume that the years to manufacture are fixed (i.e. there will not be any questions requiring you to change the number of years a vessel takes to produce).
9. The price of each vessel type is provided in the Inputs Workbook. These prices are provided in 2016 dollars and should be inflated according to the inflation instructions.
10. When a vessel is sold, the customer must pay 50% of the price in the first year. The balance due is paid in equal amounts over the remaining years of production. For example, for a Sapphire class vessel (3 year production) 50% is paid in year one and the remaining 50% is split evenly over the remaining two years (25% in year two and 25% in year three).
11. The price is fixed at the time the vessel is sold and payment allocated according to the above bullet. No further price inflation is applied to the vessel over the payment period.
12. Revenue for a vessel sale is not recognized on the Income Statement until the production is complete. For example, for a Sapphire class vessel sold in year 1, the revenue will be recognized on the Income Statement in year 3.
13. The difference between the revenue received as cash and the revenue recognized on the Income Statement will give rise to an account for revenue received in advance (unearned revenue) which must be included in any forecast Balance Sheet.
14. Cost of goods sold (COGS) for a vessel is determined as a percentage of the sale price and is provided in the Inputs Workbook. This COGS figure is spread evenly over the vessel production period. COGS is the same for cash and accrual purposes.
15. There are four other costs (head office costs, insurance, government fees and other expenses). These costs are provided in 2016 dollars and should be inflated according to the inflation instructions. These costs are the same for cash and accrual purposes and are booked in the year they occur.
16. The company is taxed at 32% of Net Profit Before Tax (NPBT). If the NPBT is negative, MOV will receive a refund. If the NPBT is positive, MOV will pay tax. There is no consideration of tax losses and tax is paid/received in the same year it is calculated for.



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### CAPITAL EXPENDITURE

17. There are three fixed asset categories: Buildings, Equipment and Other. The opening asset values, depreciation rates and forecast capital expenditure profiles are provided in the Inputs Workbook.
18. Depreciation is calculation using the diminishing value (i.e. declining balance) method. Tax and accounting depreciation are the same.
19. Local depreciation regulations mean you should assume that all capital expenditures are depreciated from the beginning of the period (i.e. depreciation is calculated on the sum of the opening value and any additional capital expenditure from that year). The capital expenditure cash flow occurs at the end of the year.
20. The capital expenditures are provided in 2016 dollars and should be inflated according to the inflation instructions.
21. Capital expenditure is fully paid in the year it is incurred. No fixed assets are disposed in the forecast period.

### FINANCING STRUCTURE

22. MOV has a complicated financing structure that will need to be carefully considered.
23. The structure includes cash, a revolver facility, two tranches of senior debt, one tranche of subordinated debt, once tranche of debt with a convertible interest stream and three classes of equity.
24. Interest for cash and interest for all debt facilities are calculated on the opening balance of the relevant facility (i.e. the prior period's closing balance). All interest rates are per annum (i.e. calculate interest as opening balance multiplied by interest rate).
25. Where there is insufficient funds from cash and the revolver facility (see below) to fund the business operations and financing requirements, the following priority order determines what should be paid first:
  - i. All revenue, COGS, other operating expenses, tax, capital expenditure, interest on cash balances, interest on Senior Debt (both tranches) and interest on the revolver facility should be paid first. These are the only items that can cause the cash balance to become negative.
  - ii. Repayments of principal on the Senior Debt. Both tranches of Senior Debt are of the same priority.
  - iii. Repayments of principal on the Subordinated Debt.
  - iv. Payments of interest on the Subordinated Debt.
  - v. Payments of interest on the Debt with Convertible Interest.
  - vi. Cash float.
  - vii. The revolver facility should be used to cover items (i) to (vi) when required.
26. The information below provides further detail on how to treat each of these financing items when insufficient funds are available.



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### CASH

27. MOV wants to have \$100m of cash available as a cash float in every year, if funds are available.
28. Positive cash balances earn interest at 2.5% p.a. Negative cash balances must pay interest at 20% p.a.
29. Where there is insufficient cash from the business operations (after tax and capital expenditure payments) to fund the financing costs, MOV will need to draw on the revolver facility (subject to the limit and other revolver information below).
30. There are certain items that must be paid out of the cash balance, even if this causes the cash balance to become negative. If the sum of the following items is negative and there are insufficient funds in the revolver to cover them, the cash balance should go negative to fund these items. These items are:
  - i. All operating flows (revenue, COGS, other operating expenses).
  - ii. Tax.
  - iii. Capital expenditure.
  - iv. Interest on the cash balance, the revolver facility and both senior debt tranches.
31. Where there is excess cash after all debt payments are made, cash float requirements are satisfied and the revolver facility is repaid, MOV will look to make a payment to shareholders (subject to the equity information below).

### REVOLVER FACILITY

32. There is a revolver facility available. It has an initial balance of zero, an interest rate of 13% per annum (p.a.) and a limit of \$5,000m.
33. If there is insufficient cash to fund the business operations or debt financing, the revolver facility should be drawn down to fund these items in accordance with the priority order provided above.
34. Once all debt payments are made and the cash float is satisfied, any residual cash should be used to repay the revolver facility.
35. The revolver facility must be fully repaid before any distributions to shareholders can be made.

### SENIOR DEBT (TRANCHES 1 & 2)

36. There are two tranches of Senior Debt (both of equal priority). Tranche 1 has an opening balance of \$8,300m and an interest rate of 4% p.a. Tranche 2 has an opening balance of \$6,200m and an interest rate of 6% p.a. No further drawdowns of these tranches are made.
37. Both tranches have a scheduled repayment profile provided in the Inputs Workbook.
38. Where there are insufficient funds to meet these scheduled principal repayments, any funds available should be split between the tranches in accordance with the relative sizes of the scheduled repayments for that year (including any deferred repayments as discussed below).
39. If the full scheduled repayment is not made, the unpaid portion must be rolled forward into the scheduled payment for the next year (and so on until it is paid). Any allocation of partial payment between the tranches should include any repayments rolled forward.



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### SUBORDINATED DEBT

40. There is one tranche of Subordinated Debt with an opening balance of \$1,800m and an interest rate of 8% p.a. No further drawdown of this tranche is made.
41. In each year a principal payment on the Subordinated Debt should be made if funds are available. This principal payment should be equal to the opening balance divided by the years remaining in the model (until 2046, including the current year).
42. If the full payment cannot be made, a reduced payment is made and this will impact the payment for the next period.
43. If the interest on the subordinated facility cannot be paid, the unpaid portion should be capitalized into the balance of the subordinated debt.
44. Any capitalized interest on the Subordinated Debt should still be included in the interest calculation on the Income Statement.

### CONVERTIBLE INTEREST DEBT

45. There is one tranche of Convertible Interest Debt with an opening balance of \$600m and an interest rate of 12% p.a.
46. No drawdowns or repayments to the Convertible Interest Debt are made.
47. If the full interest amount cannot be paid, the unpaid interest is converted into Class C equity at \$9 per share (do not round the number of shares). No inflation is applied to the conversion price per share.
48. Any interest converted into equity should still be included in the interest calculation on the Income Statement.
49. The principal of the debt does not get converted.

### EQUITY

50. There are three classes of equity – Class A (opening balance of 2,525 million shares at \$4 each), Class B (opening balance of 1,300 million shares at \$6 each) and Class C (opening balance of 700 million shares at \$9 each).
51. No new shares are issued or bought back during the period (with the exception of any Class C shares created by the Convertible Interest Debt).
52. The shares all carry different rights to any dividends payable. One B share is equivalent to 0.5 A shares. One C share is equivalent to 0.2 A shares.
53. Each class of equity cannot receive any dividends until that class of share has 'vested'. Class A vests in 2025, Class B vests in 2028 and Class C vests in 2031.
54. In addition, any new Class C shares from the Convertible Interest Debt must additionally wait 2 years from the date they convert before they can receive dividends. For example, if interest converts in 2017, it will become eligible for dividend payments at the latter of two years later (2019) or the Class C vesting year (2031) so will vest in 2031.
55. Once all operating and debt payments are made, MOV will distribute any excess cash (above the cash float) to the shareholder classes in proportion to the number and relative weighting of shares held eligible for dividends (factoring in the different rights and vesting periods listed above).



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### QUESTIONS (CIRCLE OR WRITE YOUR ANSWER)

Your Name: \_\_\_\_\_

ALL ANSWERS ARE IN UNITS OF \$m UNLESS OTHERWISE NOTED. NEGATIVE VALUES ARE GIVEN IN BRACKETS

#### Question 1

What is the total cash received from vessel sales in 2030? [6 marks total]

| Question | 1.1       | 1.2            | 1.3           |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 25,531    | 16,563         | 14,605        |
| B        | 25,532    | 16,564         | 14,606        |
| C        | 25,533    | 16,565         | 14,607        |
| D        | 25,534    | 16,566         | 14,608        |
| E        | 25,535    | 16,567         | 14,609        |
| F        | 25,536    | 16,568         | 14,610        |
| G        | 25,537    | 16,569         | 14,611        |
| H        | 25,538    | 16,570         | 14,612        |
| I        | 25,539    | 16,571         | 14,613        |
| Cell ref |           |                |               |

#### Question 2

What is the closing balance for revenue received in advance on 31 December 2032? [6 marks total]

| Question | 2.1       | 2.2            | 2.3           |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 53,281    | 27,113         | 20,278        |
| B        | 53,282    | 27,114         | 20,279        |
| C        | 53,283    | 27,115         | 20,280        |
| D        | 53,284    | 27,116         | 20,281        |
| E        | 53,285    | 27,117         | 20,282        |
| F        | 53,286    | 27,118         | 20,283        |
| G        | 53,287    | 27,119         | 20,284        |
| H        | 53,288    | 27,120         | 20,285        |
| I        | 53,289    | 27,121         | 20,286        |
| Cell ref |           |                |               |



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### Question 3

Your Name: \_\_\_\_\_

What are the total of COGS and other operating expenses in 2026? [6 marks total]

| Question | 3.1       | 3.2            | 3.3           |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 12,873    | 12,449         | 9,438         |
| B        | 12,874    | 12,450         | 9,439         |
| C        | 12,875    | 12,451         | 9,440         |
| D        | 12,876    | 12,452         | 9,441         |
| E        | 12,877    | 12,453         | 9,442         |
| F        | 12,878    | 12,454         | 9,443         |
| G        | 12,879    | 12,455         | 9,444         |
| H        | 12,880    | 12,456         | 9,445         |
| I        | 12,881    | 12,457         | 9,446         |
| Cell ref |           |                |               |

### Question 4

What is the total EBIT over the 30 year forecast period? [6 marks total]

| Question | 4.1       | 4.2            | 4.3           |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 112,830   | 18,201         | 8,790         |
| B        | 112,831   | 18,202         | 8,791         |
| C        | 112,832   | 18,203         | 8,792         |
| D        | 112,833   | 18,204         | 8,793         |
| E        | 112,834   | 18,205         | 8,794         |
| F        | 112,835   | 18,206         | 8,795         |
| G        | 112,836   | 18,207         | 8,796         |
| H        | 112,837   | 18,208         | 8,797         |
| I        | 112,838   | 18,209         | 8,798         |
| Cell ref |           |                |               |





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### Question 5

What is the total NPAT in 2017? [6 marks total]

| Question | 5.1       | 5.2            | 5.3           |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | (4,004)   | (3,837)        | (3,662)       |
| B        | (4,003)   | (3,836)        | (3,661)       |
| C        | (4,002)   | (3,835)        | (3,660)       |
| D        | (4,001)   | (3,834)        | (3,659)       |
| E        | (4,000)   | (3,833)        | (3,658)       |
| F        | (3,999)   | (3,832)        | (3,657)       |
| G        | (3,998)   | (3,831)        | (3,656)       |
| H        | (3,997)   | (3,830)        | (3,655)       |
| I        | (3,996)   | (3,829)        | (3,654)       |
| Cell ref |           |                |               |

### Question 6

What is the closing balance of the revolver facility in 2017? [6 marks total]

| Question | 6.1       | 6.2            | 6.3           |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 25        | 640            | 1,255         |
| B        | 26        | 641            | 1,256         |
| C        | 27        | 642            | 1,257         |
| D        | 28        | 643            | 1,258         |
| E        | 29        | 644            | 1,259         |
| F        | 30        | 645            | 1,260         |
| G        | 31        | 646            | 1,261         |
| H        | 32        | 647            | 1,262         |
| I        | 33        | 648            | 1,263         |
| Cell ref |           |                |               |





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### Question 7

What is the closing balance of the Senior Debt (Tranche 1 only) in 2019? [6 marks total]

| Question | 7.1       | 7.2            | 7.3           |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 7,802     | 7,886          | 7,949         |
| B        | 7,803     | 7,887          | 7,950         |
| C        | 7,804     | 7,888          | 7,951         |
| D        | 7,805     | 7,889          | 7,952         |
| E        | 7,806     | 7,890          | 7,953         |
| F        | 7,807     | 7,891          | 7,954         |
| G        | 7,808     | 7,892          | 7,955         |
| H        | 7,809     | 7,893          | 7,956         |
| I        | 7,810     | 7,894          | 7,957         |
| Cell ref |           |                |               |

### Question 8

What is the closing balance of the Subordinate Debt in 2019? [6 marks total]

| Question | 8.1       | 8.2            | 8.3           |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 1,618     | 1,813          | 1,806         |
| B        | 1,619     | 1,814          | 1,807         |
| C        | 1,620     | 1,815          | 1,808         |
| D        | 1,621     | 1,816          | 1,809         |
| E        | 1,622     | 1,817          | 1,810         |
| F        | 1,623     | 1,818          | 1,811         |
| G        | 1,624     | 1,819          | 1,812         |
| H        | 1,625     | 1,820          | 1,813         |
| I        | 1,626     | 1,821          | 1,814         |
| Cell ref |           |                |               |



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Your Name: \_\_\_\_\_

### Question 9

What is the closing cash balance in 2018? [6 marks total]

| Question | 9.1       | 9.2            | 9.3           |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 3,983     | 2,595          | 1,829         |
| B        | 3,984     | 2,596          | 1,830         |
| C        | 3,985     | 2,597          | 1,831         |
| D        | 3,986     | 2,598          | 1,832         |
| E        | 3,987     | 2,599          | 1,833         |
| F        | 3,988     | 2,600          | 1,834         |
| G        | 3,989     | 2,601          | 1,835         |
| H        | 3,990     | 2,602          | 1,836         |
| I        | 3,991     | 2,603          | 1,837         |
| Cell ref |           |                |               |

### Question 10

What is the closing balance of the Senior Debt (Tranche 2 only) in 2038? [6 marks total]

| Question | 10.1      | 10.2           | 10.3          |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 1,652     | 1,890          | 1,649         |
| B        | 1,653     | 1,891          | 1,650         |
| C        | 1,654     | 1,892          | 1,651         |
| D        | 1,655     | 1,893          | 1,652         |
| E        | 1,656     | 1,894          | 1,653         |
| F        | 1,657     | 1,895          | 1,654         |
| G        | 1,658     | 1,896          | 1,655         |
| H        | 1,659     | 1,897          | 1,656         |
| I        | 1,660     | 1,898          | 1,657         |
| Cell ref |           |                |               |



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### Question 11

Your Name: \_\_\_\_\_

What is the closing balance of the Subordinate Debt in 2038? [6 marks total]

| Question | 11.1      | 11.2           | 11.3          |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 477       | 782            | 980           |
| B        | 478       | 783            | 981           |
| C        | 479       | 784            | 982           |
| D        | 480       | 785            | 983           |
| E        | 481       | 786            | 984           |
| F        | 482       | 787            | 985           |
| G        | 483       | 788            | 986           |
| H        | 484       | 789            | 987           |
| I        | 485       | 790            | 988           |
| Cell ref |           |                |               |

### Question 12

What is the total interest paid in cash on the Convertible Interest Debt over the 30 years?  
[6 marks total]

| Question | 12.1      | 12.2           | 12.3          |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 2,155     | 1,368          | 1,797         |
| B        | 2,156     | 1,369          | 1,798         |
| C        | 2,157     | 1,370          | 1,799         |
| D        | 2,158     | 1,371          | 1,800         |
| E        | 2,159     | 1,372          | 1,801         |
| F        | 2,160     | 1,373          | 1,802         |
| G        | 2,161     | 1,374          | 1,803         |
| H        | 2,162     | 1,375          | 1,804         |
| I        | 2,163     | 1,376          | 1,805         |
| Cell ref |           |                |               |



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### Question 13

Your Name: \_\_\_\_\_

What is the sum total of the annual net drawdowns made from the revolver facility over the 30 year forecast period? Do not include any years where there was an annual net repayment in this calculation. [6 marks total]

| Question | 13.1      | 13.2           | 13.3          |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 3,451     | 15,129         | 16,464        |
| B        | 3,452     | 15,130         | 16,465        |
| C        | 3,453     | 15,131         | 16,466        |
| D        | 3,454     | 15,132         | 16,467        |
| E        | 3,455     | 15,133         | 16,468        |
| F        | 3,456     | 15,134         | 16,469        |
| G        | 3,457     | 15,135         | 16,470        |
| H        | 3,458     | 15,136         | 16,471        |
| I        | 3,459     | 15,137         | 16,472        |
| Cell ref |           |                |               |

### Question 14

What is the total number of Class C shares (in units of millions) vested at the end of 2046? [6 marks total]

| Question | 14.1      | 14.2           | 14.3          |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 700       | 765            | 734           |
| B        | 701       | 766            | 735           |
| C        | 702       | 767            | 736           |
| D        | 703       | 768            | 737           |
| E        | 704       | 769            | 738           |
| F        | 705       | 770            | 739           |
| G        | 706       | 771            | 740           |
| H        | 707       | 772            | 741           |
| I        | 708       | 773            | 742           |
| Cell ref |           |                |               |



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Your Name: \_\_\_\_\_

### Question 15

What is the sum of the total dividends paid over the 30 year forecast period to all shareholders?  
[6 marks total]

| Question | 15.1      | 15.2           | 15.3          |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 166,914   | 19,713         | 73,174        |
| B        | 166,915   | 19,714         | 73,175        |
| C        | 166,916   | 19,715         | 73,176        |
| D        | 166,917   | 19,716         | 73,177        |
| E        | 166,918   | 19,717         | 73,178        |
| F        | 166,919   | 19,718         | 73,179        |
| G        | 166,920   | 19,719         | 73,180        |
| H        | 166,921   | 19,720         | 73,181        |
| I        | 166,922   | 19,721         | 73,182        |
| Cell ref |           |                |               |

### Question 16

What is the sum of the total dividends paid over the 30 year forecast period to Class A shareholders? [6 marks total]

| Question | 16.1      | 16.2           | 16.3          |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| A        | 131,842   | 18,474         | 57,330        |
| B        | 131,843   | 18,475         | 57,331        |
| C        | 131,844   | 18,476         | 57,332        |
| D        | 131,845   | 18,477         | 57,333        |
| E        | 131,846   | 18,478         | 57,334        |
| F        | 131,847   | 18,479         | 57,335        |
| G        | 131,848   | 18,480         | 57,336        |
| H        | 131,849   | 18,481         | 57,337        |
| I        | 131,850   | 18,482         | 57,338        |
| Cell ref |           |                |               |



## Finals Section 4 - Case Study Information Pack

### Question 17

Your Name: \_\_\_\_\_

**What is the total liabilities balance at the end of 2046? Do not include any negative cash balances in your total liabilities. [6 marks total]**

| <i>Question</i> | <i>17.1</i>      | <i>17.2</i>           | <i>17.3</i>          |
|-----------------|------------------|-----------------------|----------------------|
| <b>Case</b>     | <b>Base Case</b> | <b>Declining Case</b> | <b>Cyclical Case</b> |
| <i>Marks</i>    | <i>[3 marks]</i> | <i>[2 marks]</i>      | <i>[1 mark]</i>      |
| <b>A</b>        | 107,890          | 13,907                | 89,260               |
| <b>B</b>        | 107,891          | 13,908                | 89,261               |
| <b>C</b>        | 107,892          | 13,909                | 89,262               |
| <b>D</b>        | 107,893          | 13,910                | 89,263               |
| <b>E</b>        | 107,894          | 13,911                | 89,264               |
| <b>F</b>        | 107,895          | 13,912                | 89,265               |
| <b>G</b>        | 107,896          | 13,913                | 89,266               |
| <b>H</b>        | 107,897          | 13,914                | 89,267               |
| <b>I</b>        | 107,898          | 13,915                | 89,268               |
| <b>Cell ref</b> |                  |                       |                      |



## Finals Section 4 - Case Study Information Pack

Your Name: \_\_\_\_\_

### Question 18

In how many years is the closing cash balance less than the desired cash float? [6 marks total]

| Question | 18.1      | 18.2           | 18.3          |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| Answer   |           |                |               |
| Cell ref |           |                |               |

### Question 19

If you increase the initial price of all vessels by 10%, what is the total NPAT over the 30 year forecast period? [6 marks total]

| Question | 19.1      | 19.2           | 19.3          |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| Answer   |           |                |               |
| Cell ref |           |                |               |

### Question 20

If you change the revolver limit to \$1,000m, what is the total NPAT over the 30 year forecast period? Remember to undo your sensitivity from the previous question. [6 marks total]

| Question | 20.1      | 20.2           | 20.3          |
|----------|-----------|----------------|---------------|
| Case     | Base Case | Declining Case | Cyclical Case |
| Marks    | [3 marks] | [2 marks]      | [1 mark]      |
| Answer   |           |                |               |
| Cell ref |           |                |               |

This is the end of this case study information pack.